Haesung Lee

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Summary

Computer science undergraduate interested in storage/memory systems and hardware/software co-design, with research experience in ext4 journaling. Also interested in backend engineering and blockchain.

Education

Sungkyunkwan University — Suwon, South Korea
Bachelor of Computer Science and Engineering | GPA: 4.19/4.5 (Major: 4.31/4.5) — *Mar. 2022 – Feb. 2028 (expected)*

Experience

Platform Engineer Intern — Jul. 2026 – Present
HANA Core SQL Engine 5, SAP Labs Korea Ltd. — *Seoul, South Korea*

- Contributing to the HANA native JSON column project in C++, focusing on path-index design and implementation.

Undergraduate Researcher — Mar. 2026 – Jun. 2026; Aug. 2023 – Feb. 2024
Intelligent Embedded Systems Lab., Sungkyunkwan University — *Suwon, South Korea*

- Reviewed CXL-SSD and shadow-page research, set up an emulation environment, and explored host-device interface options for metadata, device-side GC, caching, and prefetching.
- Identified lost batching and flush amplification in ext4, then implemented selective fast commit using fsync intervals and transaction dirty size to route between fast commit and jbd2.
- Measured average and maximum fsync latency across 8 KiB–4 MiB writes and 1–32 threads; identified workload-dependent gains and lock contention at high concurrency.

Software Engineer Intern — Oct. 2025 – Feb. 2026
Digital Asset, Shinhan Securities Co., Ltd. — *Seoul, South Korea*

- Decomposed a monolithic crypto trading platform into microservices.
- Rebuilt its telemetry service in Go as a non-blocking fan-out pipeline.
- Provisioned a K3s HA cluster on AWS and automated delivery with ArgoCD GitOps.

Projects

Yorkie (Wafflebase) | github.com/yorkie-team/yorkie — Jul. 2026 – Present

- Benchmarked Go 1.26 Green Tea GC against the legacy GC on Yorkie's CRDT data structures.
- Fixed several bugs in a CRDT-based real-time collaboration server and refactored its client and server components.
- Built browser-side JSON/NDJSON import for Wafflebase Sheets and revised worksheet axis updates.

Codedang | github.com/skkuding/codedang — Feb. 2026 – Present; Aug. 2022 – Feb. 2024

- Contributed 20K+ lines of code and led the backend engineering team.
- Diagnosed the root cause of database connection pool failures in the online judge and implemented a fix.
- Rebuilt the Go judge server for parallel execution and extended it into a general-purpose sandbox.
- Provided on-call support for 3+ programming contests and 6+ undergraduate courses.

Technical Skills

Languages: C/C++, Go, TypeScript/JavaScript, Python, SQL
Others: Linux, Kubernetes, Docker, AWS, gRPC, GitHub Actions, ArgoCD

Certifications, Scholarships & Others

Samsung Software Certificate Professional
Samsung Electronics — *Mar. 2026*

Digital Talent Scholarship
KT Group Hope Sharing Foundation — *Mar. 2026*

Republic of Korea Army
Sergeant, honorably discharged — *Apr. 2024 – Oct. 2025*